Loops and Iteration

CSCI 1010 demo notes - written by the Hyntor team for seed data

Why loops
A loop repeats a block of code while a condition holds. It replaces copy-pasted statements with one rule, so a change is made in one place.

while loops
A while loop checks its condition before every pass. If the condition is false at the start, the body never runs. Something inside the body must eventually make the condition false, or the loop never ends.

for loops
A for loop walks over a sequence: for item in items. Use it when you know what you are iterating over. Use range(n) to repeat n times; the counter runs from 0 to n - 1.

Off-by-one errors
The most common loop bug. Check the first and last pass by hand: does the loop touch index 0? Does it stop before len(items)? Writing the boundary cases down catches most of these.

Accumulators
To total or collect values, create a variable before the loop (total = 0, found = []) and update it on each pass. Read it after the loop ends.

break and continue
break leaves the loop immediately. continue skips to the next pass. Prefer a clear loop condition over many breaks.

Practice
1. Sum the even numbers from 1 to 100. 2. Count the vowels in a word. 3. Find the largest number in a list without using max().